

# VARDHAN-v3 Multi-Scale Dual-Domain RF Architecture Specification

Total Trainable Parameters: Exactly 119,806 | Theoretical Complexity: 11.81M MACs

## Input RF Waveform

Shape:  $(B, 1, 2048)$  @ 100 MS/s

Per-Segment Mean Detrending:  $x_d = x - \mu_x$

Per-Segment Normalization:  $(x - \mu)/\sigma$

## Temporal Backbone (MS-TCN)

Stem: Conv1D( $1 \rightarrow 32$ ,  $k=15$ ,  $s=2$ ) + BN + GELU

MS-Block 1:  $32 \rightarrow 48$ ,  $s=2$ ,  $d=1$  |  $k \in \{3, 7, 15, 31\}$

+ SE( $r=4$ ) + Residual  $\rightarrow (B, 48, 512)$

MS-Block 2:  $48 \rightarrow 64$ ,  $s=2$ ,  $d=2$  |  $k \in \{3, 7, 15, 31\}$

+ SE( $r=4$ ) + Residual  $\rightarrow (B, 64, 256)$

MS-Block 3:  $64 \rightarrow 80$ ,  $s=2$ ,  $d=4$  |  $k \in \{3, 7, 15, 31\}$

+ SE( $r=4$ ) + Residual  $\rightarrow (B, 80, 128)$

MS-Block 4:  $80 \rightarrow 80$ ,  $s=1$ ,  $d=8$  |  $k \in \{3, 7, 15, 31\}$

+ SE( $r=4$ ) + Residual  $\rightarrow (B, 80, 128)$

Attention Pooling:  $80 \rightarrow 40 \rightarrow 1$

Output Feature:  $h_T \in \mathbb{R}^{B \times 80}$

## Dual Spectral Backbone

Deterministic rFFT(2048)  $\rightarrow$  Drop DC bin  $k = 0$

3 Spectral Channels:  $(B, 3, 1024)$

Ch 0:  $P[k] = \log(1 + 1000|X[k]|^2/2048)$

Ch 1:  $R[k] = \text{Re}(X[k])/\sqrt{2048}$

Ch 2:  $I[k] = \text{Im}(X[k])/\sqrt{2048}$

Stem: Conv1D( $3 \rightarrow 32$ ,  $k=11$ ,  $s=2$ ) + BN + GELU

MS-Block 1:  $32 \rightarrow 48$ ,  $s=2$ ,  $d=1$  |  $k \in \{5, 11\}$  + SE

MS-Block 2:  $48 \rightarrow 64$ ,  $s=2$ ,  $d=2$  |  $k \in \{5, 11\}$  + SE

MS-Block 3..4:  $64 \rightarrow 80 \rightarrow 80$ ,  $d \in \{4, 8\}$  + SE

Attention Pooling:  $80 \rightarrow 40 \rightarrow 1$

Output Feature:  $h_F \in \mathbb{R}^{B \times 80}$

## Bi-Directional Cross-Domain Gating & Head

Cross Gating:  $g_T = \sigma(G_T(h_F))$ ,  $g_F = \sigma(G_F(h_T))$

Modulation:  $h'_T = h_T \odot g_T$ ,  $h'_F = h_F \odot g_F$

Fused Representation:  $h_{cat} = [h_T, h_F, h'_T, h'_F] \in \mathbb{R}^{B \times 320}$

Classifier: Linear( $320 \rightarrow 64$ ) + BN + GELU + Drop(0.2) + Linear( $64 \rightarrow 4$ )

Output: Logits  $\hat{y} \in \mathbb{R}^{B \times 4}$